

(TR-104) PYTHON WITH AI/ML

Training Day 19 Report

Date: 5 February 2026

Topic Covered: Revision of Multithreading, Multiprocessing & Performance Optimization Concepts

The nineteenth day of training focused on revising previously covered concepts of multithreading, multiprocessing, and performance optimization in Python. The day also included practical reinforcement of key modules and implementation techniques.

- Revised multiprocessing module concepts.
- Revised multithreading concepts and different ways to implement threads in Python.
- Reviewed performance optimization topics including time complexity basics.
- Practiced use of `__slots__` for memory optimization.
- Implemented and revised multiprocessing Pipe function for inter-process communication.
- Strengthened understanding of time, timeit, and cProfile modules.
- Revisited threading and multiprocessing modules through hands-on practice.
- Reinforced understanding of concurrent execution and performance improvement techniques.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/Python_Practice_Codes/tree/main/05-02-2026
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/oop-memory-optimization-in-python-using-slots-3498826e7457>